

AI Engineer – Take-Home Challenge

About Jobnova:

Jobnova is an AI agent–driven future job infrastructure that automates how people discover, match, and apply for work. We believe humans deserve better opportunities in the AI era!

AI agents help us find direct hiring signals and we update on our [Linkedin page](#).

About this take-home challenge

AI coding is very encouraged, and there are 3 parts of the challenge, you can select 1--2 that you are mostly familiar with to implement a demo.

Once you finish, send your submission and a go-through video to kevin@libaspace.com

Note: Each part is the real product feature we are going to build, so please be serious on your demo.

Part 1

AI Mock Interview demo

1. Implement the self-introduction and past-experience stages of the interview process using LiveKit AI Multi-Agent, ensuring stable performance and real time interaction with low latency in reply or follow up questions from the interviewer.
2. Ensure smooth and natural transitions between the two stages, with well-defined switching logic and no interruptions, conflicts, or repetitive prompts.
3. Provide a time-based fallback mechanism for stage transitions to guarantee continuous workflow progression when the normal switching logic is not triggered.

4. Tavus API Integration

- a. Use Tarsus Persona API ([Docs](#)) for real-time 2D/3D avatar rendering.
- b. Implement low-latency TTS and ensure accurate lip-sync.

5. Expected Outcome

- a. A working prototype of real-time digital human interaction, **must be voice interaction** with an AI avatar speaking in real time, no obvious latency/delay, gentle interruption handling should be included.
- b. Smooth logic transitioning from self-introduction stage to past experience stage.

Part 2

Linkedin Job Source agent

1. From Linkedin job listing pages, crawl the company name and company website URL(You may utilize third party Linkedin crawler API if you need)
2. From company website URL, use web agent to navigate to the career page URL
3. You should be able to get to job listing page of individual company's website
4. Test your job source agent with randomly selected 20 Linkedin job posting URLs, and report back the successful rate.
5. Example: Your agent should take this input(<https://www.linkedin.com/jobs/view/4427787182/>), and successfully output this: (https://jobs.ashbyhq.com/harvey?utm_source=58AzKxpog0)

6. Expected Outcome

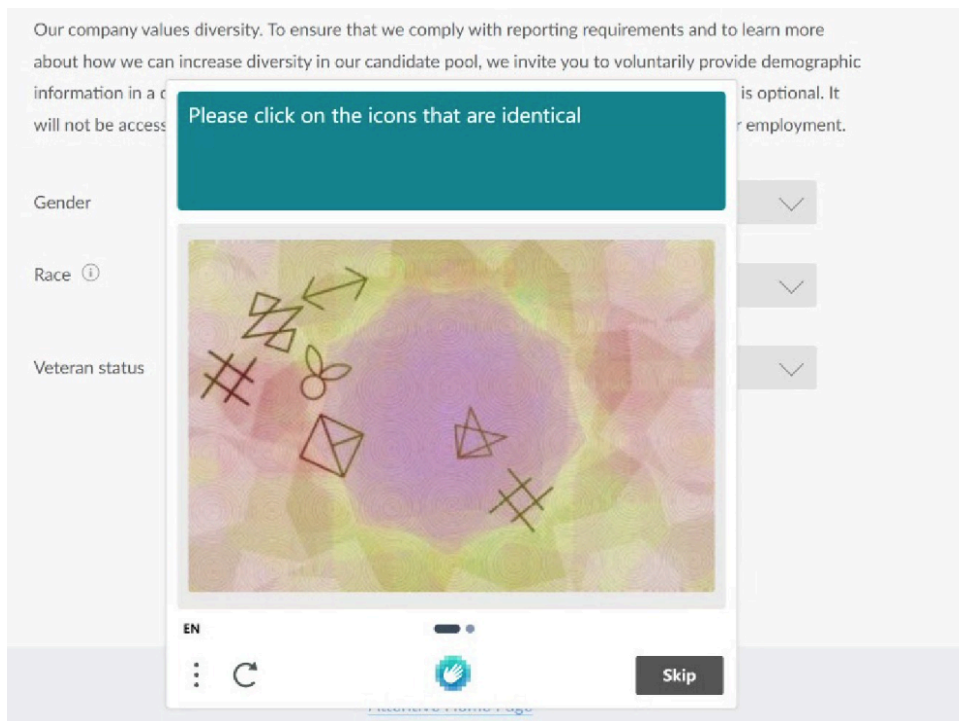
- a. The solution needs to be generic and apply widely to different formats of company websites.
- b. Test your job source agent with randomly selected 20 Linkedin job posting URLs, and report back the successful rate for getting the jobs list page of the company website. Show me the results in your demo video.
- c. Attached a deployed website URL, so we can also randomly select 10 linkedin URLs to test out your solution.

Part 3

Job auto apply agent

1. While submitting job applications is time consuming, we are developing a job auto apply agent for ATS based websites like greenhouse workday Glassdoor ... etc.
2. For now we use playwright to fill the application forms in the headless chrome in EC2 server. But we will meet human verification when the playwright tries to click the submit button for some website, sometimes like workday, lever.
3. We would need to build a chrome plug-in, and run it purely on server side, so it can work around with this.
4. Your task: Work on the following job application from Lever, that can auto apply without the blocking issue(consider chrome extension to automate):

<https://jobs.lever.co/ekimetrics/d9d64766-3d42-4ba9-94d4-f74cdaf20065>



☒ YES ☐ NO

* Are you located in and eligible to work in Canada (Citizen, Permanent Resident, Work Permit)?

☒ YES ☐ NO



Verify you are human



Cancel

Submit application